STATISTICS

PART 7: Sampling Theory (for Two Populations)

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Sampling Distribution for Two Populations

The comparison of parameters between two population constitutes an important class of problems in statistics. In this context, the following parameters will be analyzed:

- the difference between population means → $\mu_1 - \mu_2$
- the difference between population proportions → $p_1 - p_2$
- The ratio of population variances → $\dfrac{\sigma_1^2}{\sigma_2^2}$

Sampling Theory for the Difference between Means

Consider a sample of size n_1 whose members are

$$X_{11}, X_{12}, \dots, X_{1n_1}$$

is selected from a population with mean μ_1 and variance σ_1^2, and similarly another sample of size n_2 whose members are

$$X_{21}, X_{22}, \dots, X_{2n_2}$$

is selected from a population with mean μ_2 and variance σ_2^2. Then,

$$E(\bar{X}_1 - \bar{X}_2) = E(\bar{X}_1) - E(\bar{X}_2) = \mu_1 - \mu_2,$$

$$\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = (1)^2\mathrm{Var}(\bar{X}_1) + (-1)^2\mathrm{Var}(\bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

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Theorem: If the two populations are distributed normally with the means μ_1 and μ_2, and the variances σ_1^2 and σ_2^2, then the random variable

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a standard normal distribution, where n_1 and n_2 are sample sizes.

Proof: Since $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$,

$$\bar{X}_1 \sim N\left(\mu_1, {}^{\sigma_1^2}/_{n_1}\right) \text{ and } \bar{X}_2 \sim N\left(\mu_2, {}^{\sigma_2^2}/_{n_2}\right),$$

then we have

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, {}^{\sigma_1^2}/_{n_1} + {}^{\sigma_2^2}/_{n_2}\right)$$

$$\Rightarrow \quad z = \frac{(\bar{X}_1 - \bar{X}_2) - E(\bar{X}_1 - \bar{X}_2)}{\sqrt{\sigma^2_{\bar{X}_1 - \bar{X}_2}}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Sampling Theory for the Difference between Means (for unknown variances)

Theorem: If the two populations are distributed normally with the means μ_1 and μ_2, and their variances are equal, i.e., $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (but their values are unknown), then the random variable

$$t_{n_1+n_2-2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_k\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a t distribution with degree of freedom $n_1 + n_2 - 2$. Here,

$$S_k = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}.$$

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Proof: Since $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$, the random variables

$$\chi_1{}^2 = \frac{(n_1-1)S_1^2}{\sigma_1^2} \text{ and } \chi_2{}^2 = \frac{(n_2-1)S_2^2}{\sigma_2^2}$$

have Chi-Square distribution with the degrees of freedom $n_1 - 1$ and $n_2 - 1$. If we take $\sigma_1^2 = \sigma_2^2 = \sigma^2$, then the random variable

$$Y = \chi_1{}^2 + \chi_2{}^2 = \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} = \frac{(n_1+n_2-2)S_k^2}{\sigma^2}$$

has also Chi-Square distribution with the degrees of freedom $n_1 + n_2 - 2$ where

$$S_k = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}}.$$

Finally, from $t_v \sim \dfrac{z}{\sqrt{\chi_v^2/_v}}$,

$$\frac{z}{\sqrt{\chi^2_{n_1+n_2-2}/_{(n_1+n_2-2)}}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma^2}{n_1} + \dfrac{\sigma^2}{n_2}}} \sqrt{\frac{(n_1+n_2-2)\sigma^2}{(n_1+n_2-2)S_k^2}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_k\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$$

$\sim t_{n_1+n_2-2}$

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Theorem: If the two populations are distributed normally with the means μ_1 and μ_2, and their variances are not equal, i.e., $\sigma_1^2 \neq \sigma_2^2$ (but their values are unknown), then the random variable

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

has a t distribution with degree of freedom ν which can computed by rounding of

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}}.$$

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Sampling Theory for the Difference between Means (for unknown populations)

According to the Central Limit Theorem; if two populations has any distribution with the means μ_1 and μ_2, and the variances σ_1^2 and σ_2^2, then the random variable

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has the standard normal distribution for $n_1 \geq 30$ and $n_2 \geq 30$. Also, if the population variances σ_1^2 and σ_2^2 are unknown, then the random variable

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

has approximately the standard normal distribution for $n_1 \geq 30$ and $n_2 \geq 30$.

Sampling Theory for the Difference between Means (For Paired Populations)

Up to this point, we have investigated the difference between the means of two independent populations. However, when the measurements are paired (or repeated), the populations become dependent.

For example:

- The weight of people before and after a diet program
- The level of a blood parameter before and after taking a new drug
- The prices of a group of some specific products sold in two different shopping centers

In cases of dependent populations, such as the examples above, we should use a different technique to estimate the difference between population means.

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For paired populations:

Let n paired members be chosen from two dependent populations with random variables X_1, X_2 and means μ_1, μ_2. Then,

$$D_i = X_{1i} - X_{2i}, \qquad \text{i} = 1,2,3, \dots, \text{N}$$

is a new random variable that represents of difference between paired members, where N is the size of both populations . Thus, the population mean will be

$$\mu_D = \frac{\sum_{i=1}^{N} D_i}{N}$$

and the sample mean and sample standard deviation will be

$$\bar{D} = \frac{\sum_{i=1}^{n} D_i}{n} \qquad S_D = \sqrt{\frac{\sum_{i=1}^{n} (D_i - \bar{D})^2}{n-1}}$$

Moreover

$$E(\bar{D}) = E\left(\frac{\sum_{i=1}^{n} D_i}{n}\right) = \frac{1}{n}\sum_{i=1}^{n} E(D_i) = \frac{1}{n} n\mu_D = \mu_D$$

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Theorem: Let a sample of size n is chosen from two paired populations which are distributed normally and μ_D is the population mean of $D_i = X_{1i} - X_{2i}$. If $\overline{D}$ and S_D are the sample mean and sample standard deviation, respectively, then the random variable

$$t_{n-1} = \frac{\overline{D} - \mu_D}{S_D \big/ \sqrt{n}}$$

has a t distribution with the degree of freedom $n-1$.

According to the Central Limit Theorem; for paired populations having unknown distribution, $\overline{D}$ has normal distribution and the random variable

$$z = \frac{\overline{D} - \mu_D}{S_D \big/ \sqrt{n}}$$

has the standard normal distribution when $n \geq 30$.

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Sampling Theory for the Difference between Proportions

Let two samples with size n_1 and n_2 are from two populations, respectively. Let X_1 and X_2 be random variables related to these two populations.

If $\hat{p}_1 = \dfrac{X_1}{n_1}$ and $\hat{p}_2 = \dfrac{X_2}{n_2}$, then

$$\mathrm{E}[\hat{p}_1 - \hat{p}_2] = \mathrm{E}[\hat{p}_1] - \mathrm{E}[\hat{p}_2] = p_1 - p_2$$

and

$$\mathrm{Var}[\hat{p}_1 - \hat{p}_2] = (1)^2\mathrm{Var}(\hat{p}_1) + (-1)^2\mathrm{Var}(\hat{p}_2) = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}.$$

Sampling Theory

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According to the Central Limit Theorem; $(\hat{p}_1 - \hat{p}_2)$ has normal distribution and the random variable

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - \mathrm{E}[\hat{p}_1 - \hat{p}_2]}{\sqrt{\mathrm{Var}[\hat{p}_1 - \hat{p}_2]}} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$$

has the standard normal distribution for $n_1 \geq 30$ and $n_2 \geq 30$.

In practice, for the large values of n_1 and n_2, we can use $\hat{p}_1$ and $\hat{p}_2$ instead of p_1 and p_2. Thus, the random variable

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}}$$

has also approximately distributed standard normally.

Sampling Theory for the Ratio of Two Variances

Theorem: If S_1^2 and S_2^2 are the variances of the independent random samples of size n_1 and n_2 which are selected from two populations distributed normally with variances σ_1^2 and σ_2^2, the random variable

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = \frac{S_1^2 \sigma_2^2}{S_2^2 \sigma_1^2}$$

has a F distribution with the degree of freedoms $n_1 - 1$ and $n_2 - 1$.

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Proof:

We know that $F_{v_1,v_2} \sim \frac{U/_{v_1}}{V/_{v_2}}$ if $U \sim \chi^2_{v_1}$ and $\mathrm{V} \sim \chi^2_{v_2}$ are independent variables.

Also, since populations have normal distributions, we can write:

$$\chi^2_{n_1-1} \sim \frac{(n_1-1)S_1^2}{\sigma_1^2} \text{ and } \chi^2_{n_2-1} \sim \frac{(n_2-1)S_2^2}{\sigma_2^2}.$$

$$\Rightarrow \quad F_{n_1-1,\, n_2-1} \sim \frac{\chi^2_{n_1-1}\Big/(n_1-1)}{\chi^2_{n_2-1}\Big/(n_2-1)} = \frac{\dfrac{(n_1-1)S_1^2}{\sigma_1^2(n_1-1)}}{\dfrac{(n_2-1)S_2^2}{\sigma_2^2(n_2-1)}} = \frac{\dfrac{(n_1-1)S_1^2}{\sigma_1^2(n_1-1)}}{\dfrac{(n_2-1)S_2^2}{\sigma_2^2(n_2-1)}} = \frac{S_1^2\Big/\sigma_1^2}{S_2^2\Big/\sigma_2^2}$$

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OUTLINE: Theorems for Two Independent Populations

for population means $(\mu_1 - \mu_2)$

- $X_1 \sim N$ and $X_2 \sim N$
  - known $({\sigma_1}^2, {\sigma_2}^2)$

    $$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}$$

  - unknown $({\sigma_1}^2, {\sigma_2}^2)$
    - ${\sigma_1}^2 = {\sigma_2}^2$

      $$t_{n_1+n_2-2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_k\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

      $$\left({S_k}^2 = \frac{(n_1 - 1){S_1}^2 + (n_2 - 1){S_2}^2}{n_1 + n_2 - 2}\right)$$

    - ${\sigma_1}^2 \neq {\sigma_2}^2$

      $$t_v = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}}$$

      $$\left(v = \frac{\left(\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}\right)^2}{\frac{({s_1}^2/n_1)^2}{n_1 - 1} + \frac{({s_2}^2/n_2)^2}{n_2 - 1}}\right)$$

- unknown populations and $n_1 \geq 30, n_1 \geq 30$
  - known $({\sigma_1}^2, {\sigma_2}^2)$

    $$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}$$

  - unknown $({\sigma_1}^2, {\sigma_2}^2)$

    $$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}}$$

for difference proportions $(P_1 - P_2)$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$$

for rate of population variances $({\sigma_1}^2/{\sigma_2}^2)$

$$F_{n_1-1,n_2-1} = \frac{{S_1}^2/{\sigma_1}^2}{{S_2}^2/{\sigma_2}^2}$$